

nary way of regulating the dose after the first in practice in any event, that is, when the vaccines are standardized as to numbers.

CASE 1.—The patient, a university graduate and a high-school teacher, aged 23, had had "hay-fever" regularly the past five or six years. It began usually about the end of July, and lasted until two or three weeks after a heavy frost. July 30, 1911, 1.6 c.c. of vaccine were injected. No local or constitutional effect from the injection was noted. August 6 the patient reported that he had not sneezed since the vaccination, and nasal discharge had been practically absent; 1 c.c. of the original vaccine was injected as a second dose. During the following week the patient went on duty from 6 a. m. until 11 a. m. daily, as ticket-seller at the local driving park. The air in the ticket-booth was foul from its incidental use as a privy, and the food furnished was so poor that it could not be eaten, so that the patient said he "lived on watermelon and buttermilk." He had attacks of sneezing each morning, though little during the day. August 14, only 0.2 c.c. of a new vaccine was injected but with no noticeable effect. August 18, 0.6 c.c. of the same vaccine used which had a slight effect. August 22, 1 c.c. was injected by the patient's uncle, a physician, which gave relief for twenty-four hours only. August 28, 1 c.c. of a new vaccine was given and the patient removed to a city on Lake Erie to begin high-school work. He was convinced that he had been better than for any of the past five years. He reported in the fall of 1912 that for the past summer (not under treatment) his attacks had been scarcely one third as bad as normal.

CASE 2.—The patient, a university professor, had typical "hay-fever" for the past five or six years. It began usually about the middle of August, though this year he had been sneezing since July 1. Aug. 10, 1911, 1.3 c.c. of vaccine was injected. The dose was too large, and the reaction was severe and kept the patient in bed until the next afternoon. He did not sneeze once until the 13th. August 14, he reported that he had sneezed five or six times on the 13th and 14th, but "nose had not run since the 10th." August 16, a second vaccine was made and only 0.3 c.c. injected; the effect was only slight, and 0.4 c.c. was given August 18. From this time on the patient had so little trouble that he did not consider further vaccination necessary. He "enjoyed being in the dust where buildings were being razed [he is an architect], and sitting among the vines on his porch," which had been impossible before owing to the paroxysms. During the summer of 1912, he did not suffer enough from hay-fever to believe it worth while to be vaccinated.

CASE 3.—The patient, an agent, had hay-fever for twelve years. For the past five years he had gone to Duluth to remain from August to December to escape, though in 1911 he had a slight attack while there. Aug. 12, 1912, 0.7 c.c. was given with practically no results. August 16, 1.2 c.c. was given with practically no results up to August 27, when 1.2 c.c. of a new vaccine was given. August 31, the patient reported a good effect to date and little sneezing. September 12, 1 c.c. was given. The patient reported no hay-fever since the last vaccine. No further vaccination was given. October 12, he reported that he had no more hay-fever.

CASE 4.—The patient, a lawyer, had had hay-fever for eighteen years, and asthma the past few years with it. Aug. 13, 1912, 0.8 c.c. was given with no reaction, and no sneezing up to August 16, when the second dose, 1.2 c.c., was given. August 30, the patient reported great relief the past two weeks, with practically no hay-fever. There was no effect on the asthma. August 31, 1 c.c. of new vaccine was given and none since, as he has had no hay-fever.

CASE 5.—The patient, a business man, had had hay-fever for years. Aug. 25, 1912, 0.4 c.c., a small preliminary dose, was given, with slight effect. August 29, 1 + c.c. was given. The reaction was severe, so that the patient felt "out of sorts" for about two days; but the nose cleared up and all symptoms disappeared, and the patient had no further hay-fever, as he reported in November.

CASE 6.—The patient, a lumber-dealer, had hay-fever for five or six years. Aug. 31, 1912, 1 c.c. was given and the patient left the city. Vaccine was sent to his physician and he administered the second dose, September 6. I have only the patient's statements as to the effects. "The first vaccination (Saturday, August 31) was followed by 2 degrees of fever and a chill that evening, a slight one the following evening. Felt like taking the grip. Feverish Sunday, but better Monday. Next three days felt about like one with a cold. Second vaccination followed by much the same effects, but not so pronounced. About like one with a cold the next week, sneezed a little and some nasal discharge, but not the 'hot kind.' Felt better than since had hay-fever, about six years."

CASE 7.—The patient, a bookseller, had had hay-fever for thirty-three years. Sept. 3, 1912, 1.2 c.c. of vaccine was given. The patient reported that the nose had dried by the following day and his "necessary handkerchiefs had dropped from about twelve per day to none" for hay-fever purposes. September 7, the second and final vaccination was given, as the patient had no further attacks.

CASE 8.—The patient, a Cornell University student, aged 21, had had hay-fever during the past twelve years, and asthma with it the past three years. It lasts from about the second week in August to the end of September. Sept. 11, 1912, the first vaccine was given, and material was sent to the patient, who was given a second injection, September 17. A letter from the patient dated the 17th reported the "head fairly clear, and felt much better," but no further information was received from him or his physician.

The cases are few, but are certainly interesting, especially Cases 3, 4, 5 and 7, in which the patients were under my personal observation in 1912, as were the patients in Cases 1 and 2 in 1911. The satisfying results of these two led to the continuation of the work in 1912, and the experimental vaccination was an excellent guide to procedure. It appears of promise, also, that these two were much better in 1912, though they took no treatment.

I have made no attempt to discover any "specific germ" of hay-fever and do not believe that it exists. The theory on which the work was based was the strengthening of the nasal mucosa, so that the irritants, whatever they are, would be without effect.

THE CEREBRAL FORM OF PERNICIOUS MALARIA *

J. F. PATTERSON, M.D.

NEWBERN, N. C.

Cerebral malaria is a form of pernicious malaria, the etiologic factor being the estivo-autumnal parasite, that is, the crescent-forming hemameba. The fevers caused by the estivo-autumnal parasite are characterized by some main group of symptoms depending on which organ of the body participates mostly in the infection, in other words, the organ that is affected; for the malarial parasite of this type can attack any organ in the body and give rise to symptoms characteristic of involvement of that particular organ; for example, if the liver is involved we have bilious symptoms, if the brain, cerebral symptoms, and so on. This is the most important point to remember, for we can readily see that pernicious malaria may simulate any disease and lends emphasis to the necessity of a blood examination for the malarial parasite in all cases in which the diagnosis is obscure.

* Chairman's article before the Section of Practice of Medicine at the 1913 Session of the Medical Society of the State of North Carolina, held at Morehead City, N. C., June, 1913.